

Unit 3, Lesson 5: Using Operations to Solve Real-World Problems – Part 2

Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Learning Target

- ☐ I can solve one-step, real-world problems involving any of the four operations with decimals or fractions.

3.5.1 Warm-Up: Two Questions and One Wonder

Ky-ree, a firefighter in Los Angeles, budgets his paycheck.

Ky-ree's Budget	
Monthly salary	\$5,800
Taxes and other deductions	\$2,900
Net take home income	\$2,900
Rent	\$1,350
Utilities	\$60
Internet & cable	\$50
Cell phone	\$130
Car payment	\$330
Insurance	\$120
Gas	\$140
Food	\$250
Gym	\$30
Entertainment	\$120
Remaining	\$320

1. List two questions you have about Ky-ree's budget.
2. List one thing you wonder about Ky-ree's budget.

3.5.2 Guided Instruction: Real-World Contexts About Finances

Income is the amount of money received.

1. Recall the problem from Lesson 4.

A Valencia orange tree that is 12 years old can produce 1.4 boxes of fruit. In May 2017, a box of fresh Valencia oranges cost \$20.45. How much income would a Valencia orange tree generate?

Finance is the management of money.

2. Daniel owns a Valencia orange grove. Choose other items from the list that he should consider as a part of his finances.
- ☐ The taxes he pays on the property.
 - ☐ The amount of money he has in the bank.
 - ☐ The cost of taking care of the orange grove.
3. Use the words in the word bank to complete each statement.

Word Bank

withdrawal

checking

costs

deposit

expenses

savings

taxes

- a. _____ or _____ describe money that is spent.
- b. _____ are the amount of money that is paid to the government.
- c. A _____ is the amount of money placed in a bank account.

- d. A _____ is the amount of money removed from a bank account.
- e. _____ and _____ are types of bank accounts.

An **investment** is when the initial amount of money, known as the **principal**, earns more money, known as the **return**, over time.

4. For each financial transaction, write the mathematical operation that is most commonly associated with it.

Financial Transaction	Mathematical Operation
costs	
deposit	
expenses	
income	
taxes	
withdrawal	
return on an investment	

Gross income is the amount of income that is earned.

5. Hannah earns \$9.17 an hour. Each week she works 14 hours. How much is Hannah’s weekly gross income?

Net income is the amount of income that is earned after expenses and taxes are deducted.

6. Each week Hannah has to pay \$26.96 in taxes. What is Hannah’s net income?

3.5.3 Guided Practice: Solving Real-World Finance Problems

1. A trampoline park charges \$26.88 for 2 hours.
 - a. How much is the cost for 1 hour?
 - b. The manager of the trampoline park knows that during a Saturday there are typically 125 people at the park. If each person only pays for 2 hours, what is the park's typical income on Saturdays?
 - c. The manager of the trampoline park lists the costs for Saturday as follows: \$119.50 to pay the employees and \$256.73 for all other costs. What are the park's typical costs?
 - d. Determine the park's net income for a typical Saturday.
2. Emmanuel invested \$12,500 into the stock market. In the first quarter of the year his return was \$281.25. He combines the return with his initial investment. How much does Emmanuel have invested after the first quarter of the year?
3. Aaron has \$2,478.23 in his checking account. He writes a check for \$637.89 to pay for new tires on his truck. How much money does he have in his checking account now?
4. Violeta has \$5,734.49 in her savings account. She deposits \$218.56. How much does she have in her savings account now?

3.5.4 Wrap-Up: Ticket Out the Door

Mishka is planning to host a music festival to raise funds for a local food bank. Her budget is shown.

Item	Amount
Ticket price at the door	\$9.00
Fee paid to each band	\$175.00
Early-bird ticket price	\$6.50
Equipment rental	\$225.00
Publicity (flyers/artwork)	\$64.57
Stage rental	\$187.50

1. In order to find the **gross income** for the music festival, I need to know _____ and _____.
2. List the items in Mishka's budget that are **expenses**.
3. If Mishka had a gross income of \$1,166.50 and expenses totaling \$827.07, what is her net income for the music festival?